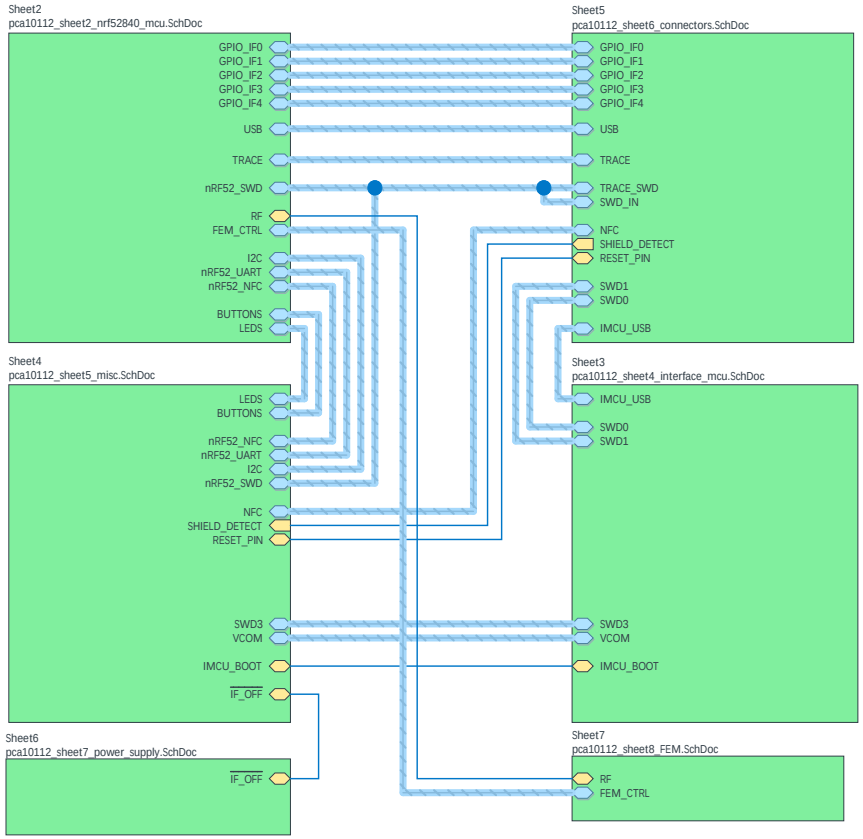
Nordic Semiconductor ASA

nRF21540 Development Kit (PCA10112)

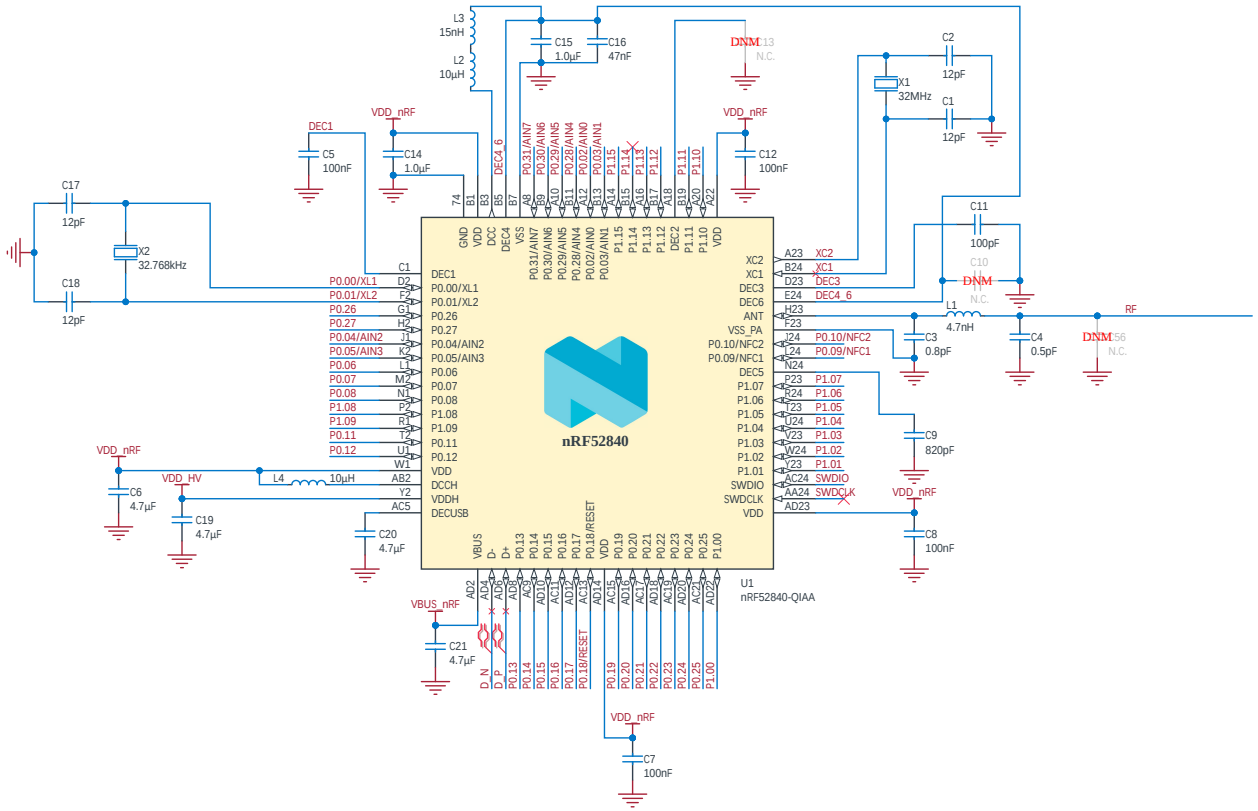
Sheet 1: Cover
Sheet 2: nRF Radio
Sheet 3: Interface MCU
Sheet 4: Miscellaneous
Sheet 5: Connectors
Sheet 6: Power Supply
Sheet 7: Front-End Module



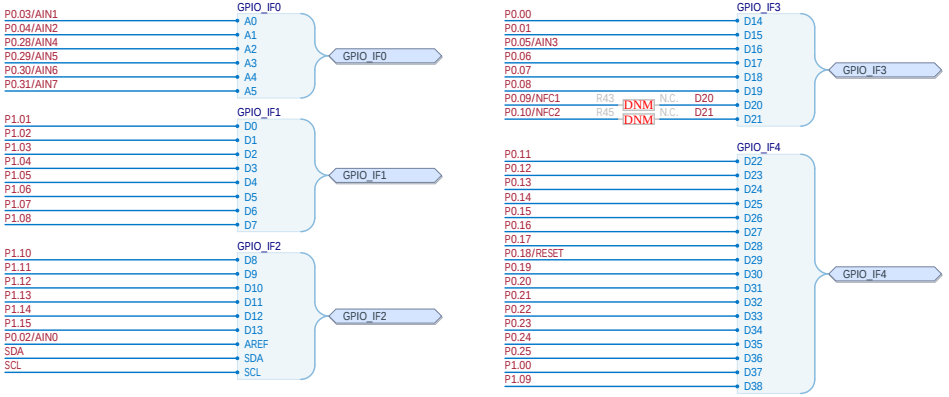
× The No ERC object is a design directive.
This directive is placed on a node in the circuit to suppress harmless warnings and/or erro r violation conditions that are detected when the schematic project is compiled.

Title: Cover		
Size: A3	PCB Assembly Number: PCA10112	Revision: 2.0.0
Date: 17.08.2022		Sheet 1 of 7
File: pca10112_sheet1_cover.SchDoc		Drawn By: MASI
Classification: Public		

NORDIC SEMICONDUCTOR



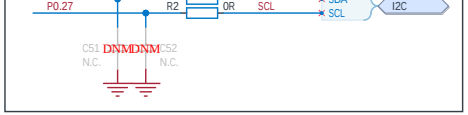
GPIO interface



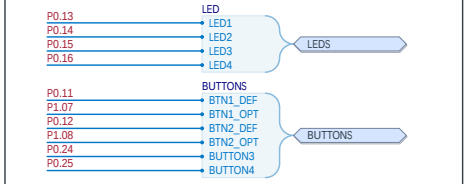
USB



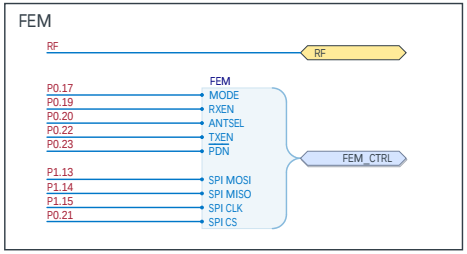
I2C



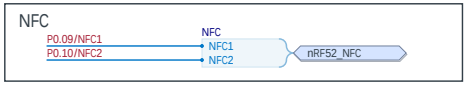
User interface



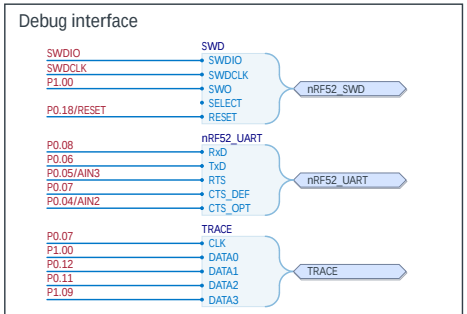
FEM



NFC

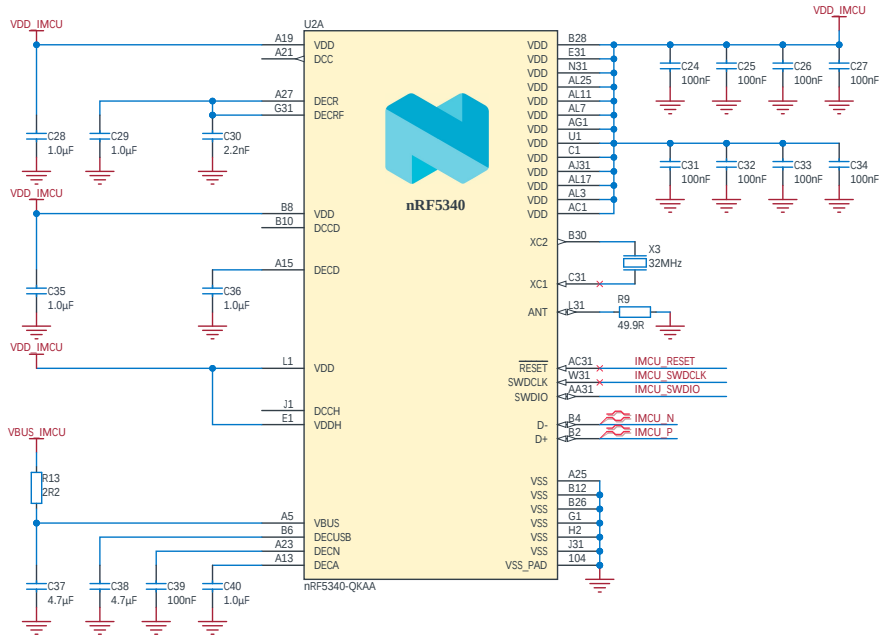


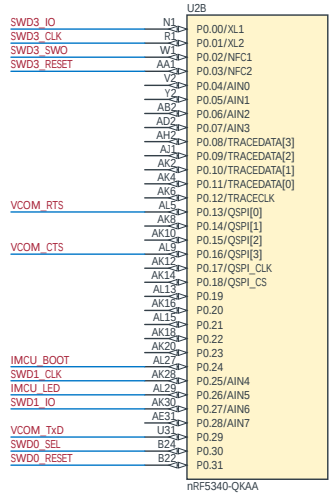
Debug interface

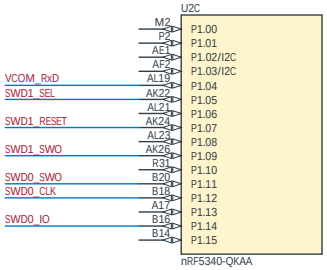


Title: nRF52840 MCU		
Size: A3	PCB Assembly Number: PCA10112	Revision: 2.0.0
Date: 17.08.2022		Sheet 2 of 7
File: pca10112_sheet2_nrf52840_mcu.SchDoc		Drawn By: MASI
Classification: Public		

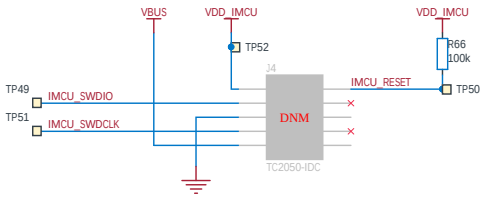
Interface MCU



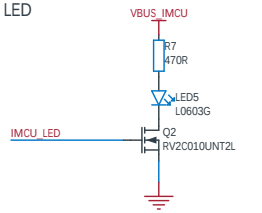




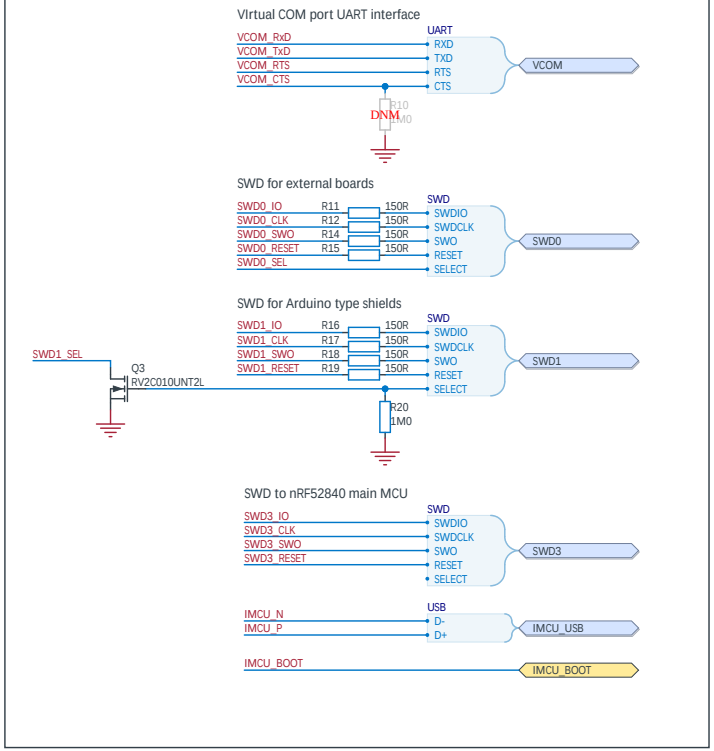
Interface MCU Programming Connector



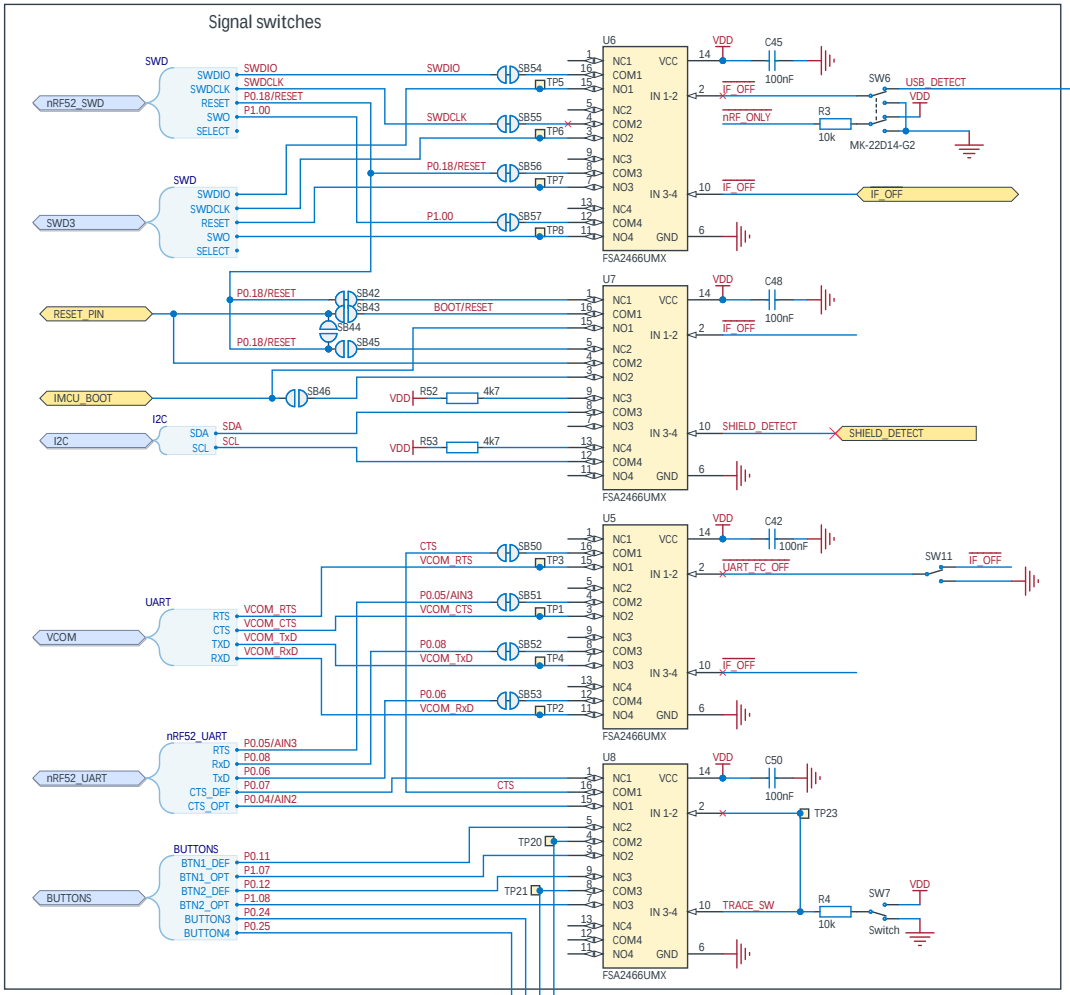
LED



Ports

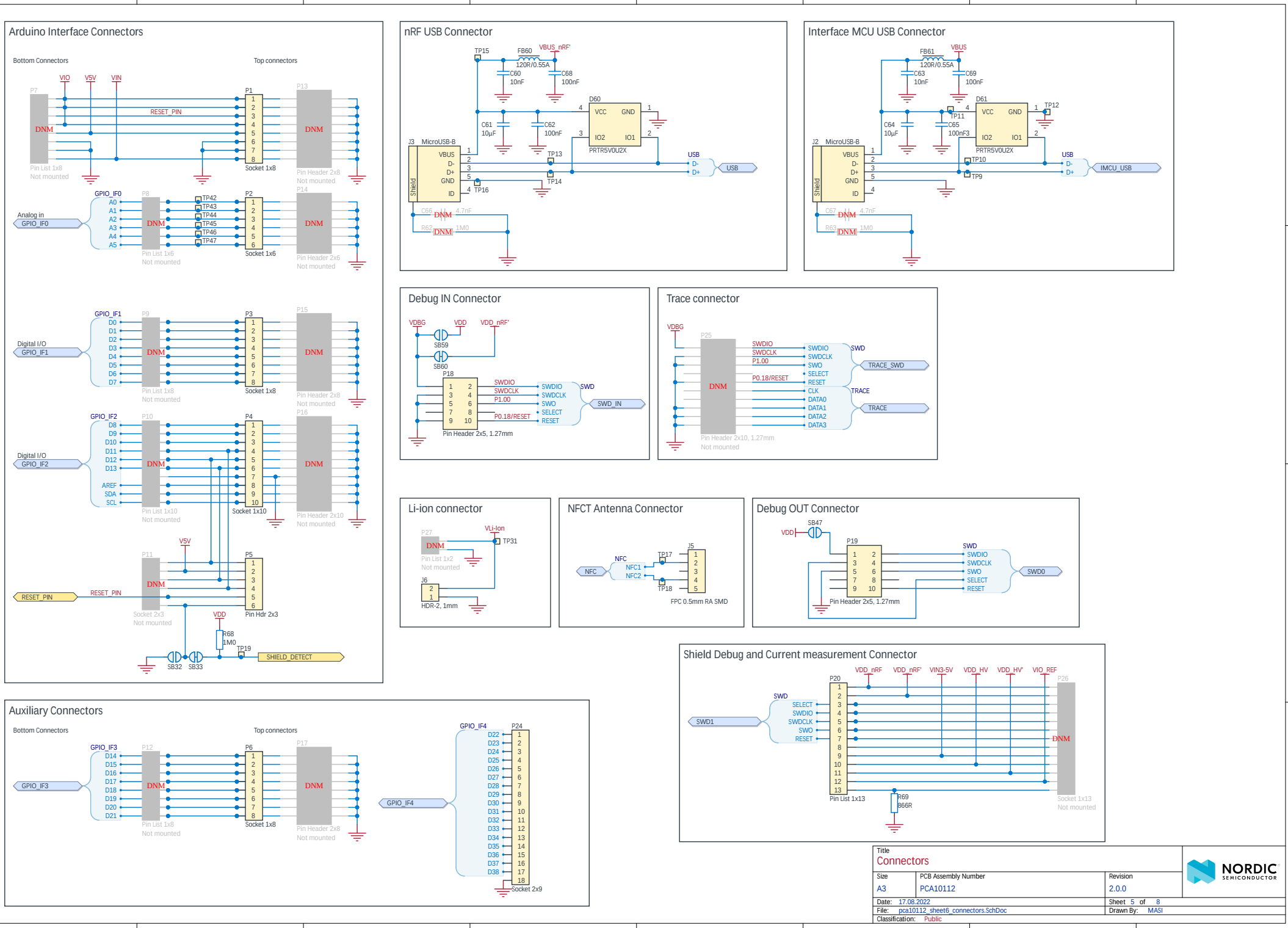


Title	Interface MCU		
Size	PCB Assembly Number	Revision	
A3	PCA10112	2.0.0	
Date:	13.10.2022	Sheet 3 of 7	
File:	pca10112_sheet4_interface_mcu.SchDoc	Drawn By: MASI	
Classification:	Public		

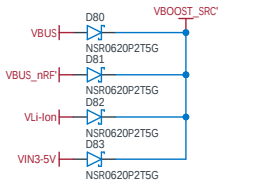


Title: Miscellaneous		
Size: A3	PCB Assembly Number: PCA10112	Revision: 2.0.0
Date: 17.08.2022		Sheet 4 of 8
File: pca10112_sheet5_misc.SchDoc		Drawn By: MASI
Classification: Public		

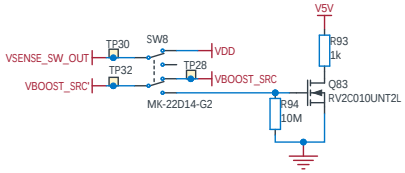
NORDIC SEMICONDUCTOR


Arduino Interface Connectors
Bottom Connectors
Top connectors
VIO
V5V
VIN
RESET_PIN
P7
DNM
P1
Socket 1x8
P13
Pin Header 2x8
Not mounted
Pin List 1x8
Not mounted
GPIO_IF0
Analog in
GPIO_IF0
A0
A1
A2
A3
A4
A5
P8
TP42
TP43
TP44
TP45
TP46
TP47
P2
Socket 1x6
P14
Pin Header 2x6
Not mounted
Pin List 1x6
Not mounted
GPIO_IF1
Digital I/O
GPIO_IF1
D0
D1
D2
D3
D4
D5
D6
D7
P9
P3
Socket 1x8
P15
Pin Header 2x8
Not mounted
Pin List 1x8
Not mounted
GPIO_IF2
Digital I/O
GPIO_IF2
D8
D9
D10
D11
D12
D13
AREF
SDA
SCL
P10
P4
Socket 1x10
P16
Pin Header 2x10
Not mounted
Pin List 1x10
Not mounted
V5V
P11
P5
Pin Hdr 2x3
Socket 2x3
Not mounted
RESET_PIN
RESET_PIN
VDD
R68
1M0
TP19
SB32
SB33
SHIELD_DETECT
nRF USB Connector
TP15
FB60
VBUS_nRF'
120R/0.55A
C60
10nF
C68
100nF
C61
10µF
C62
100nF
D60
VCC
GND
IO2
IO1
PRTR5V0U2X
J3
MicroUSB-B
VBUS
D-
D+
GND
ID
Shield
TP13
TP14
TP16
USB
USB
C66
DNM
4.7nF
R62
DNM
1M0
Interface MCU USB Connector
FB61
VBUS
120R/0.55A
C63
10nF
C69
100nF
C64
10µF
C65
100nF3
D61
TP11
TP12
TP10
TP9
J2
MicroUSB-B
IMCU_USB
C67
DNM
4.7nF
R63
DNM
1M0
Debug IN Connector
VDBG
VDD
VDD_nRF'
SB59
SB60
P18
SWDIO
SWDCLK
P1.00
P0.18/RESET
SWO
SELECT
RESET
SWD
SWD_IN
Pin Header 2x5, 1.27mm
Trace connector
VDBG
P25
DNM
SWDIO
SWDCLK
P1.00
P0.18/RESET
SWO
SELECT
RESET
CLK
DATA0
DATA1
DATA2
DATA3
SWD
TRACE_SWD
TRACE
TRACE
Pin Header 2x10, 1.27mm
Not mounted
Li-ion connector
P27
DNM
Pin List 1x2
Not mounted
VLi-Ion
TP31
J6
HDR-2, 1mm
NFCT Antenna Connector
NFC
NFC
NFC1
NFC2
TP17
TP18
J5
FPC 0.5mm RA SMD
Debug OUT Connector
VDD
SB47
P19
SWDIO
SWDCLK
SWO
SELECT
RESET
SWD
SWD0
Pin Header 2x5, 1.27mm
Shield Debug and Current measurement Connector
VDD_nRF
VDD_nRF'
VIN3-5V
VDD_HV
VDD_HV'
VIO_REF
P20
SWD
SELECT
SWDIO
SWDCLK
SWO
RESET
SWD1
DNM
P26
Pin List 1x13
Socket 1x13
Not mounted
R69
866R
Auxiliary Connectors
Bottom Connectors
Top connectors
GPIO_IF3
GPIO_IF3
D14
D15
D16
D17
D18
D19
D20
D21
P12
P6
Socket 1x8
P17
Pin Header 2x8
Not mounted
Pin List 1x8
Not mounted
GPIO_IF4
GPIO_IF4
D22
D23
D24
D25
D26
D27
D28
D29
D30
D31
D32
D33
D34
D35
D36
D37
D38
P24
Socket 2x9
Title
Connectors
Size
A3
PCB Assembly Number
PCA10112
Revision
2.0.0
Date: 17.08.2022
Sheet 5 of 8
File: pca10112_sheet6_connectors.SchDoc
Drawn By: MASI
Classification: Public
NORDIC SEMICONDUCTOR

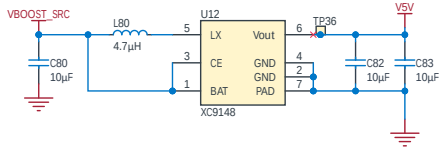
Power input, reverse protection



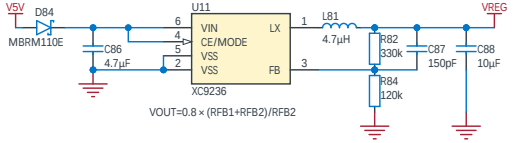
Power switch



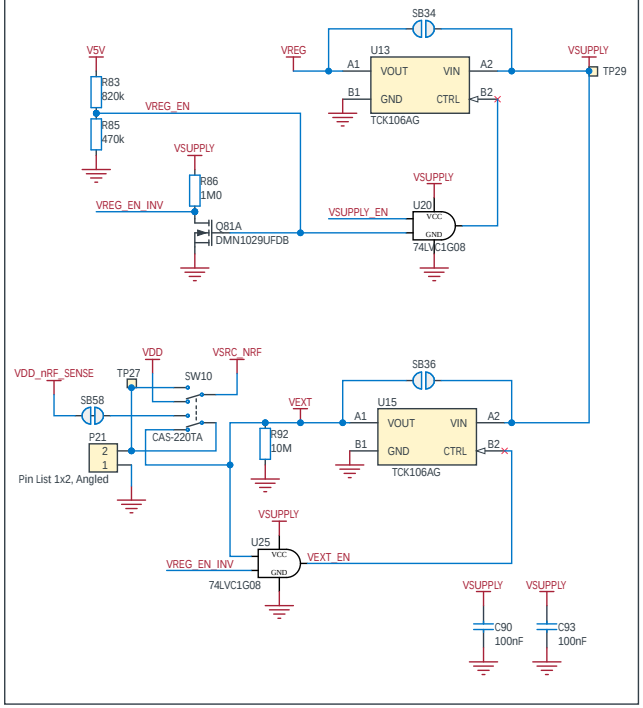
5V Boost regulator



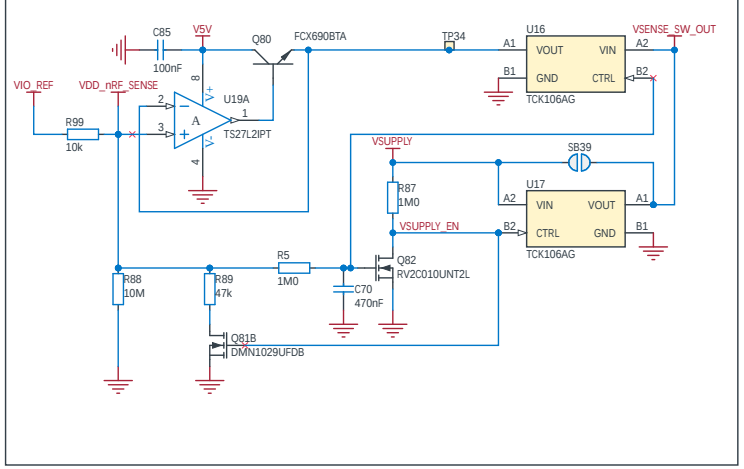
Buck regulator for VDD



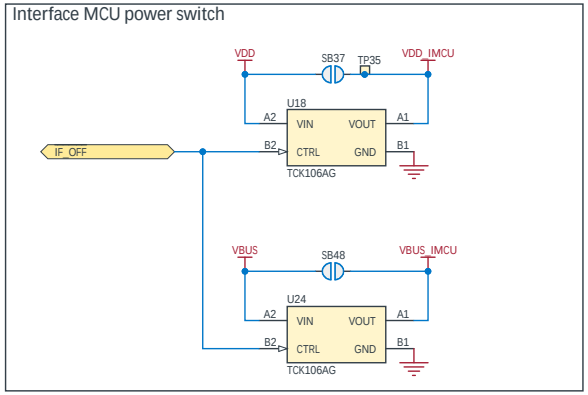
Power sources and switches



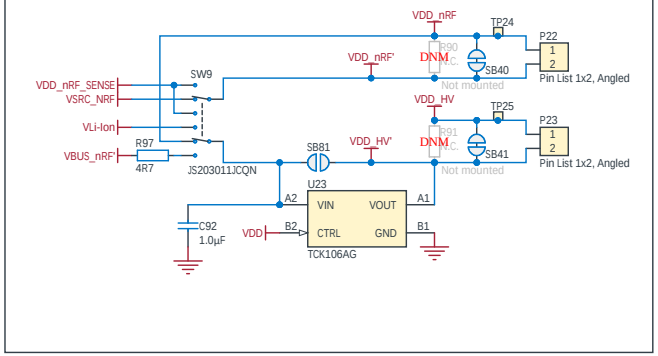
VDD_nRF sense buffer and switch



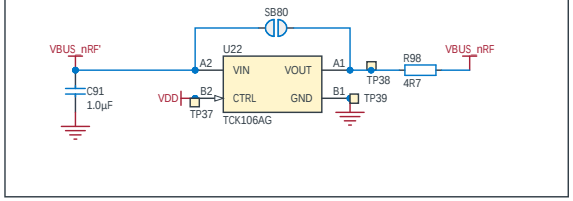
Interface MCU power switch



nRF power source select and current measurement



VBUS_nRF switch

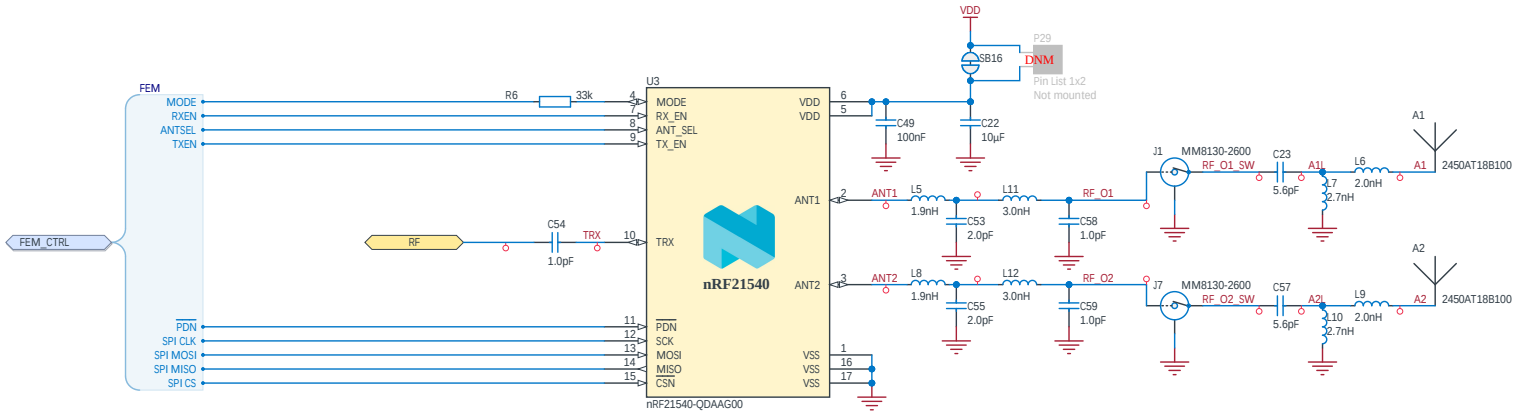


Instrument ground



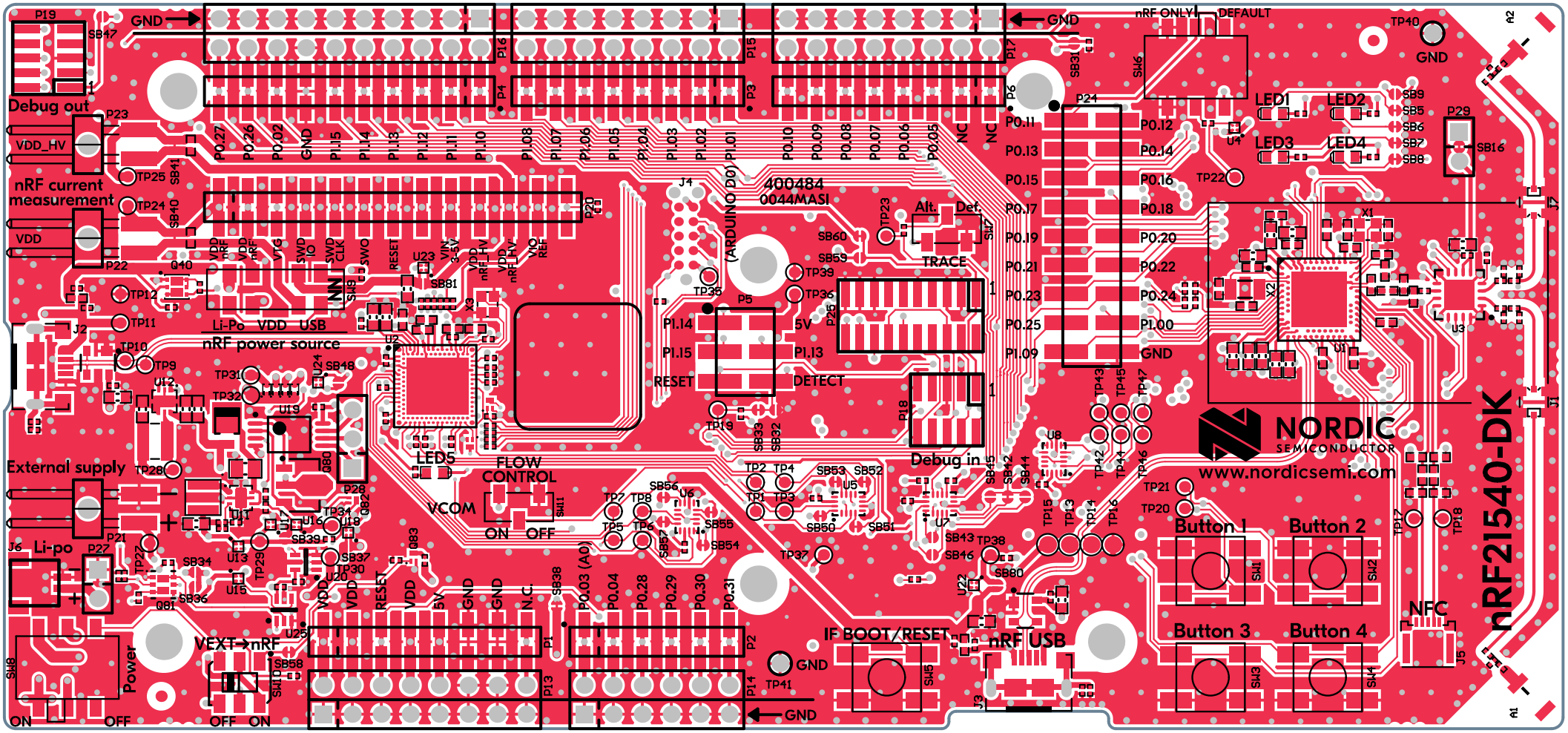
Title: Power Supply		
Size: A3	PCB Assembly Number: PCA10112	Revision: 2.0.0
Date: 17.08.2022		Sheet 6 of 7
File: pca10112_sheet7_power_supply.SchDoc		Drawn By: MASI
Classification: Public		

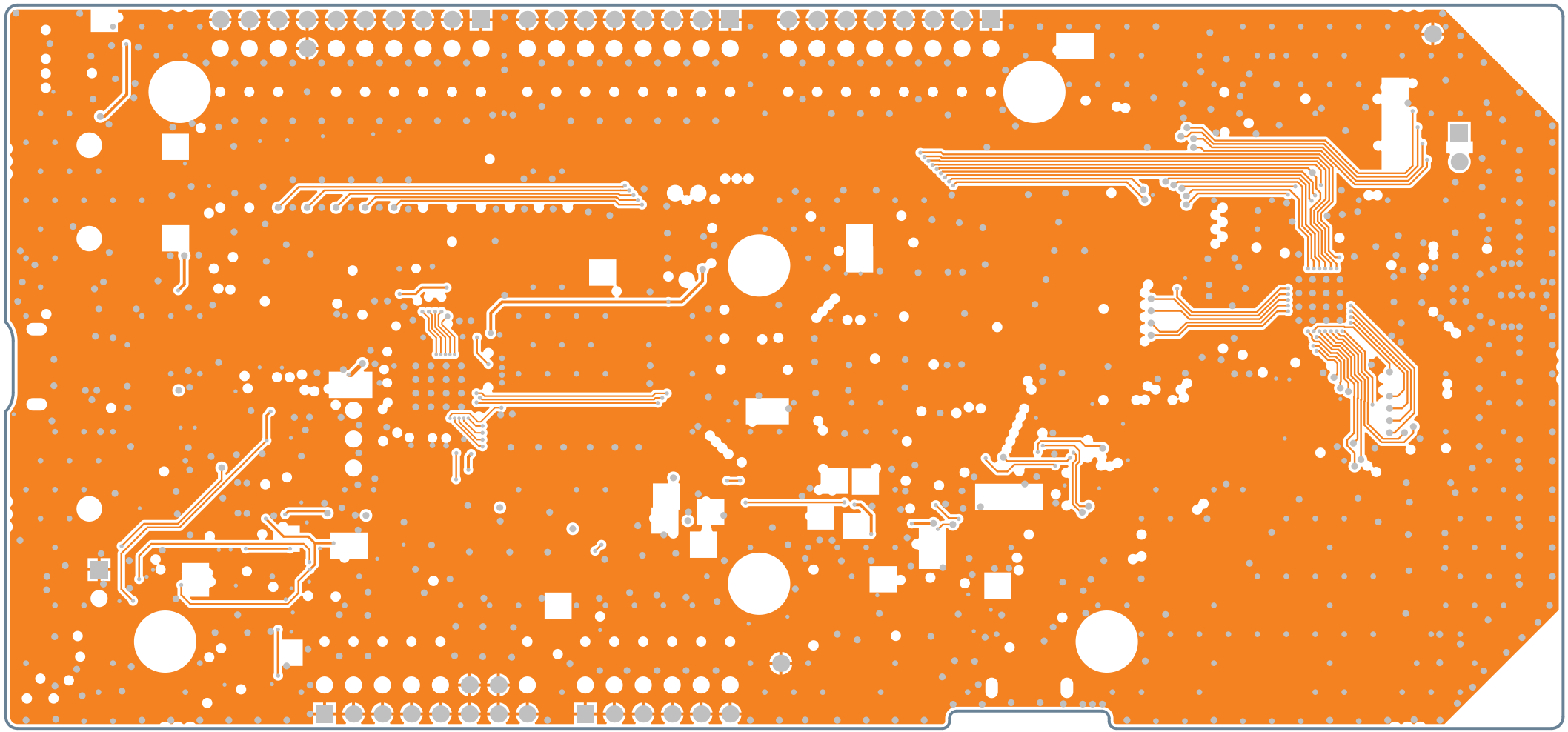
NORDIC SEMICONDUCTOR

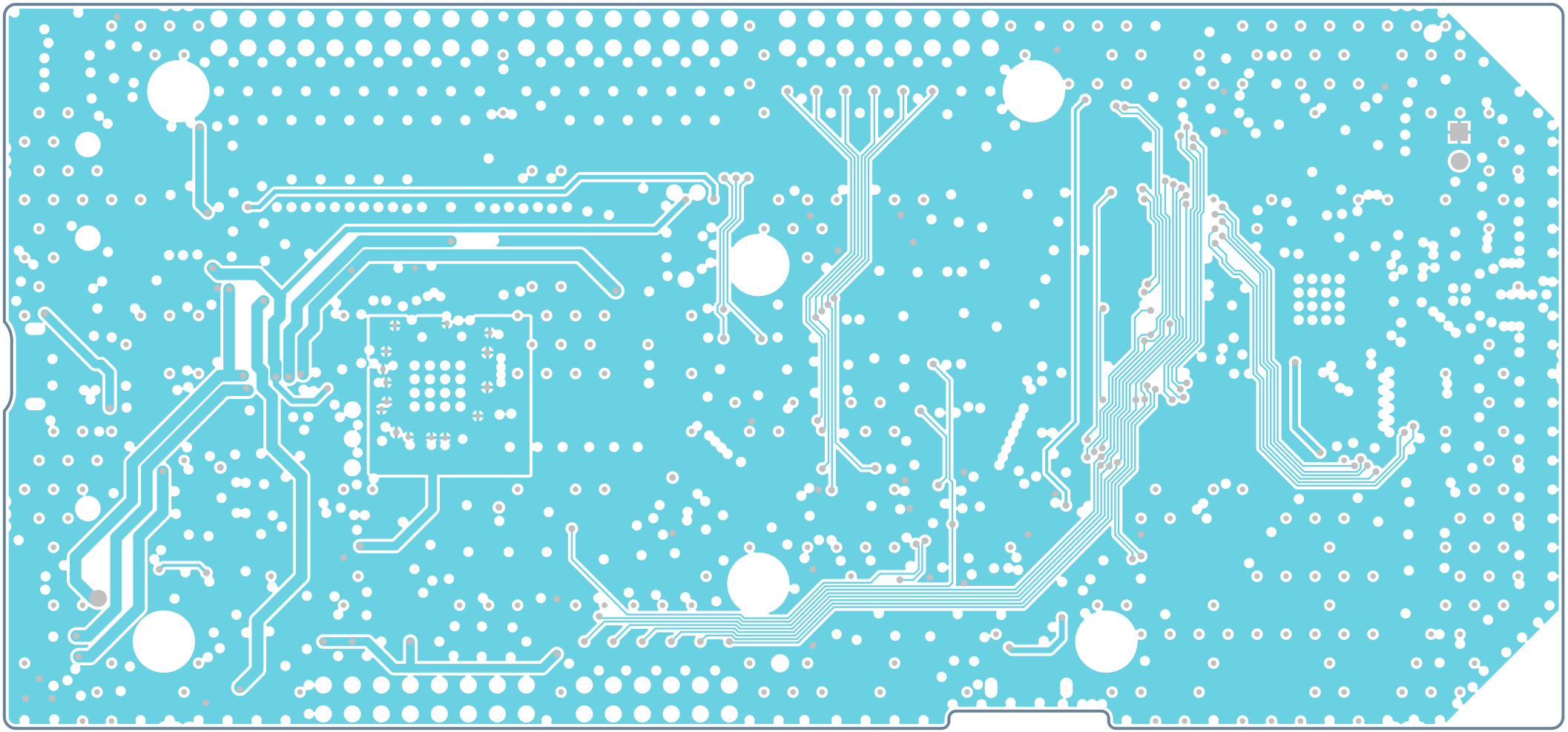


Title: Front End Module		
Size: A3	PCB Assembly Number: PCA10112	Revision: 2.0.0
Date: 17.08.2022		Sheet 7 of 7
File: pca10112_sheet8_FEM.SchDoc		Drawn By: MASI
Classification: Public		

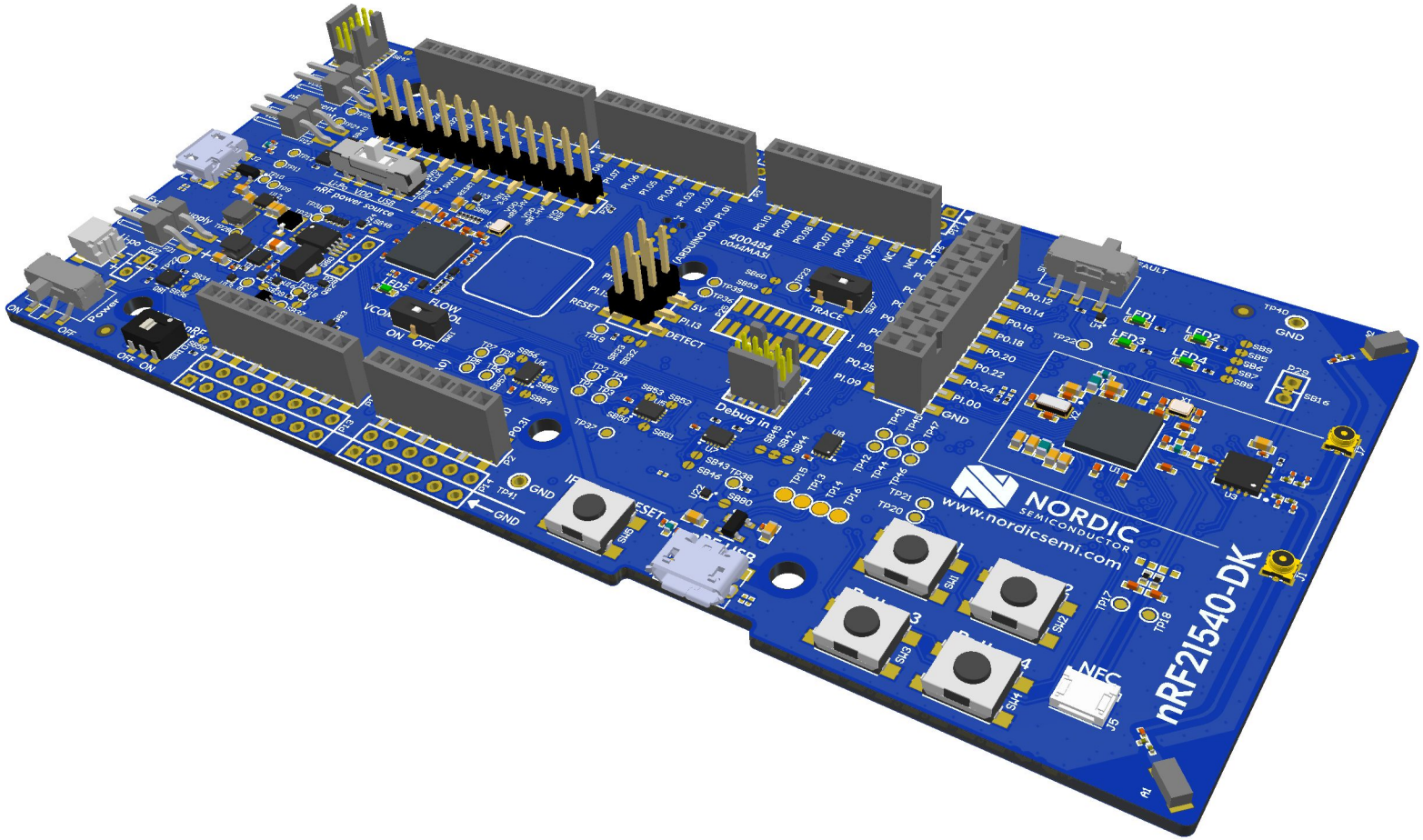
NORDIC SEMICONDUCTOR

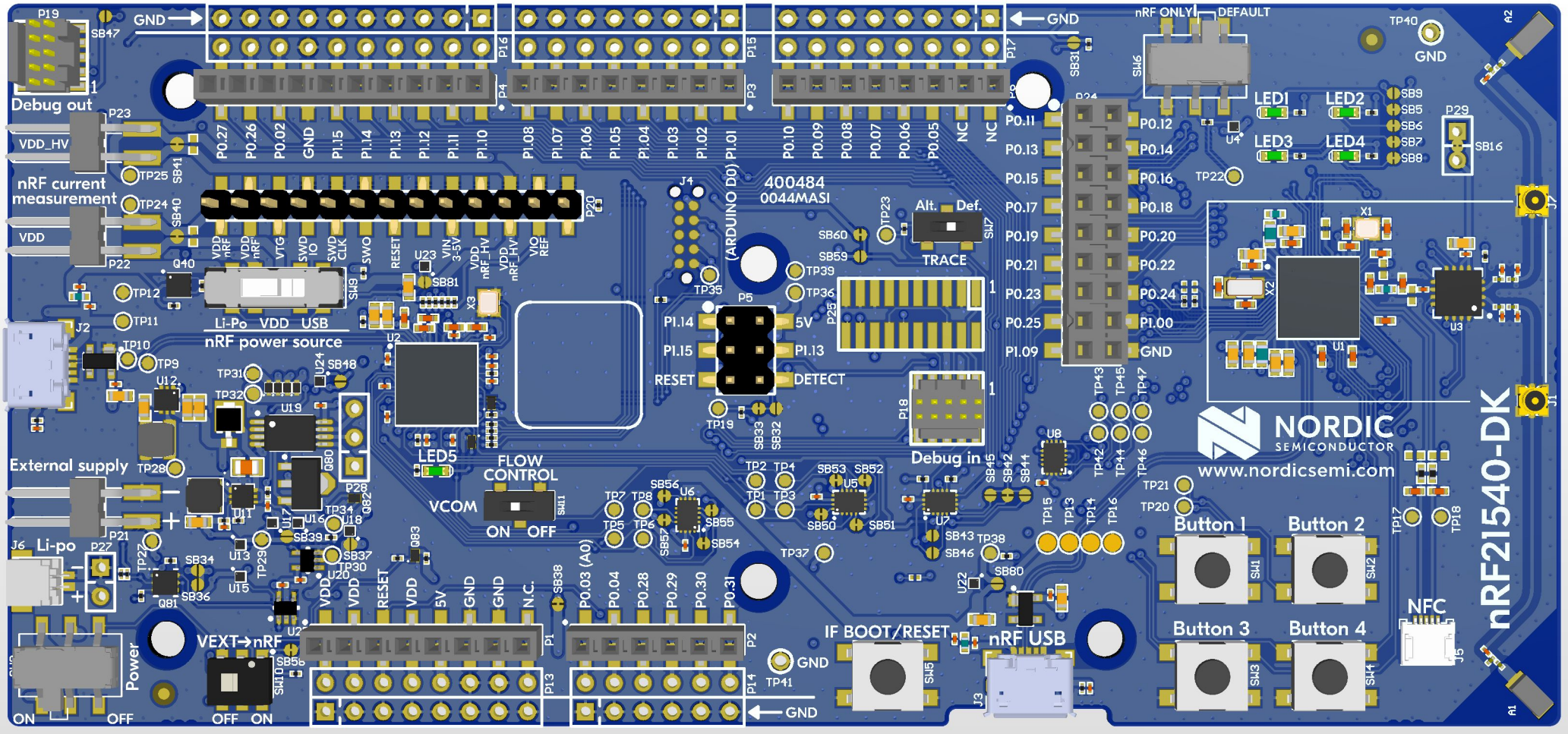

nRF21540-DK
NORDIC SEMICONDUCTOR
www.nordicsemi.com
Debug out
nRF current measurement
Li-Po VDD USB
nRF power source
External supply
Li-po
Power
VEXT→nRF
LED5
FLOW CONTROL
VCOM
ON OFF
TRACE
Alt.
Det.
Debug in
IF BOOT/RESET
nRF USB
nRF ONLY!
DEFAULT
LED1
LED2
LED3
LED4
Button 1
Button 2
Button 3
Button 4
NFC
RESET
DETECT
5V
GND
400484
0044MASI

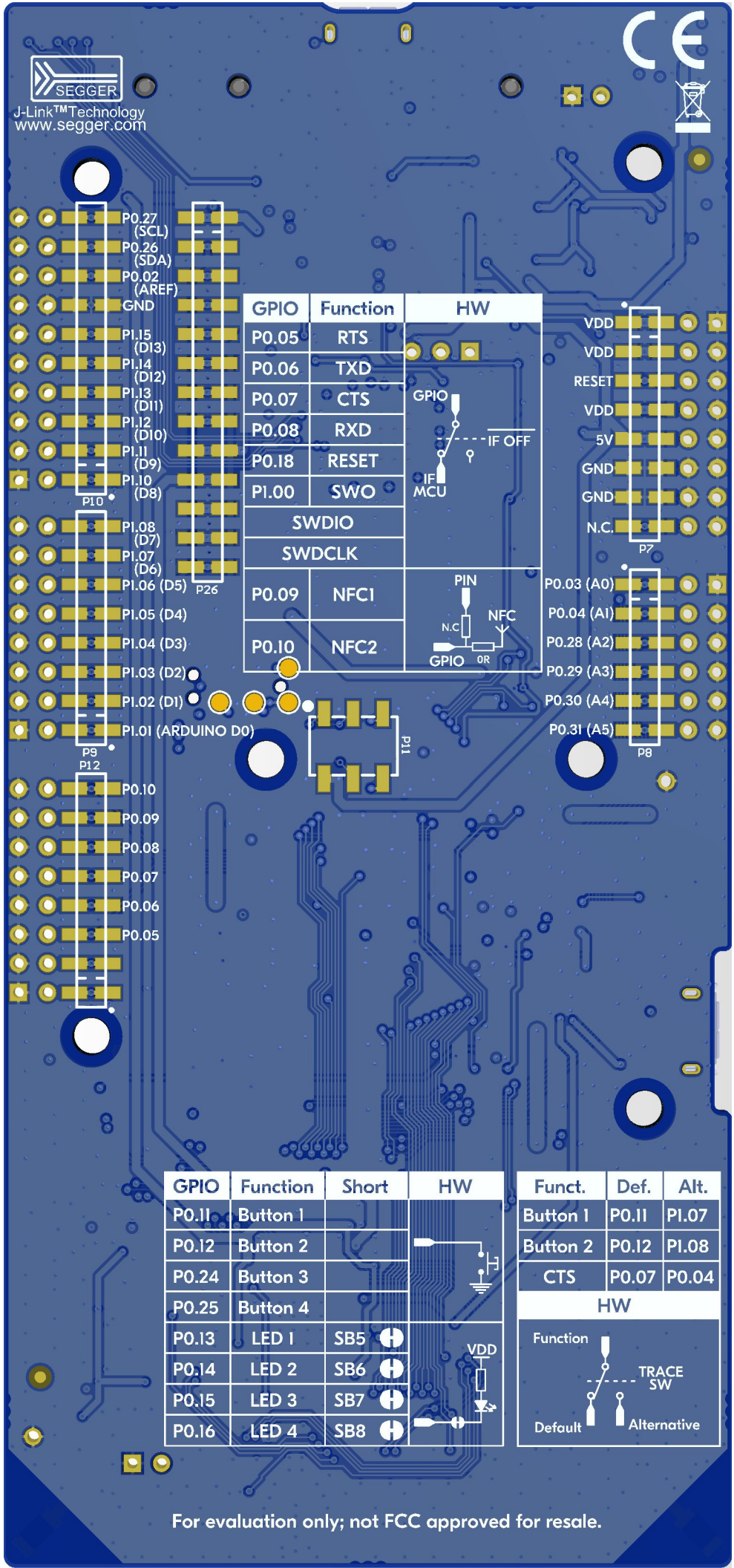




For evaluation only. not FCC approved for resale.
HW
Short
Function
GPIO
LED 4
LED 3
LED 2
LED 1
Button 4
Button 3
Button 2
Button 1
P0.04
P0.05
P0.06
P0.07
P0.08
P0.09
P0.10
P0.11
P0.12
P0.13
P1.07
P1.08
P0.28 (A3)
P0.29 (A4)
P0.30 (A5)
P0.31 (A6)
P0.02 (A0)
P0.03 (A1)
P0.04 (A2)
P0.05 (A7)
P0.06
P0.07
P0.08
P0.09
P0.10
P0.11
P0.12
P0.13
P1.01
P1.02
P1.03
P1.04
P1.05
P1.06
P1.07
P1.08
P1.09
P1.10
P1.11
P1.12
P1.13
P1.14
P1.15
P0.00
P0.01
P0.02
P0.03
P0.04
P0.05


NORDIC
SEMICONDUCTOR
www.nordicsemi.com
nRF21540-DK
NFC
GND
RESET
TRACE
DETECT
FLOW
VCOM
Debug in
nRF power source
Power
ON
OFF


nRF21540-DK
NORDIC SEMICONDUCTOR
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400484
0044MASI
Debug out
nRF current measurement
VDD
VDD_HV
nRF power source
Li-Po VDD USB
External supply
Li-po
Power
ON OFF
OFF ON
VEXT→nRF
LED5
FLOW CONTROL
VCOM
ON OFF
Debug in
TRACE
Alt. Def.
DETECT
IF BOOT/RESET
nRF USB
NFC
nRF ONLY DEFAULT
LED1
LED2
LED3
LED4
Button 1
Button 2
Button 3
Button 4
(ARDUINO D0)
(A0)
GND
5V
RESET


SEGGER
J-Link™Technology
www.segger.com
P0.27 (SCL)
P0.26 (SDA)
P0.02 (AREF)
GND
P1.15 (D13)
P1.14 (D12)
P1.13 (D11)
P1.12 (D10)
P1.11 (D9)
P1.10 (D8)
P10
P1.08 (D7)
P1.07 (D6)
P1.06 (D5)
P26
P1.05 (D4)
P1.04 (D3)
P1.03 (D2)
P1.02 (D1)
P1.01 (ARDUINO D0)
P9
P12
P0.10
P0.09
P0.08
P0.07
P0.06
P0.05
GPIO | Function | HW
P0.05 | RTS
P0.06 | TXD
P0.07 | CTS
P0.08 | RXD
P0.18 | RESET
P1.00 | SWO
SWDIO
SWDCLK
P0.09 | NFC1
P0.10 | NFC2
GPIO
IF OFF
IF MCU
PIN
N.C
NFC
GPIO
0R
P11
VDD
VDD
RESET
VDD
5V
GND
GND
N.C.
P7
P0.03 (A0)
P0.04 (A1)
P0.28 (A2)
P0.29 (A3)
P0.30 (A4)
P0.31 (A5)
P8
GPIO | Function | Short | HW
P0.11 | Button 1
P0.12 | Button 2
P0.24 | Button 3
P0.25 | Button 4
P0.13 | LED 1 | SB5
P0.14 | LED 2 | SB6
P0.15 | LED 3 | SB7
P0.16 | LED 4 | SB8
VDD
Funct. | Def. | Alt.
Button 1 | P0.11 | P1.07
Button 2 | P0.12 | P1.08
CTS | P0.07 | P0.04
HW
Function
TRACE SW
Default
Alternative
For evaluation only; not FCC approved for resale.